



UNIVERSITY
OF TRENTO

Motivations and Introduction

Databases course

Data is important

- We live in the information age
- Every minute of the day, here's how much data is generated online:
 - <https://minuteinternet.com/en/>
- According to an estimation, a typical individual creates between 50,000 to 100,000 data points daily.
 - <https://www.linkedin.com/pulse/how-much-data-can-one-person-generate-michael-jimmink-mba-dxsc>

Data is important

- Data constitutes the basic infrastructure of our lives, making things simpler, possible, better (and worse)
- Examples
 - Knowing when is the next bus is
 - Knowing in which room the DB lesson is
 - Knowing whether an invoice was paid
 - Being able to retrieve the results of my latest health check
 - Knowing ...

Data is important

- **Informed Decision-Making:**
 - Data allows to see what actually works
 - Data allows to see the effects of policies and actions
- **Personalization and Convenience:**
 - Algorithms use data to tailor experiences (for instance: movie recommendations, fastest route to avoid traffic)
- **Scientific Discovery, Problem Solving, Healthcare:**
 - Some disciplines rely heavily on data (track diseases, develop new medicines, model climate change)
- **Efficiency and Automation:**
 - Data helps streamline operations, optimize supply chains, and automate repetitive tasks, saving time and resources across industries.

Example: Managing a Company

- Information one needs:
 - ...
 -

Example: Managing a Company

- Areas to take care of to manage a company:
 - Client and Prospects
 - Orders and Sales
 - Inventory, Production,
 - Quotations, Invoicing, Accounting
 - Projects and performance
 - Human Resources, Presence, Timesheets
 - Knowledge & Documents
- Approaches:
 - On people's computers
 - On shared folders
 - On a shared system

Data is important: Example

- Profit & Loss projection
- Systems performance tracking
- Projection of inventory lock stock alerts
- User engagement and experience, Retention metrics
- Which products actually work, which functions actually work

Data is important to make informed decisions

- 2006, Clive Humby: “**Data is the new oil**”

- Raw data has little value by itself
- Refining unlocks insights

- From *data* to *structured information*

- We need a “refinery”

Leaders | Regulating the internet giants

The world’s most valuable resource is no longer oil, but data

The data economy demands a new approach to antitrust rules



May 6th 2017

Share

<https://www.economist.com/leaders/2017/05/06/the-worlds-most-valuable-resource-is-no-longer-oil-but-data>

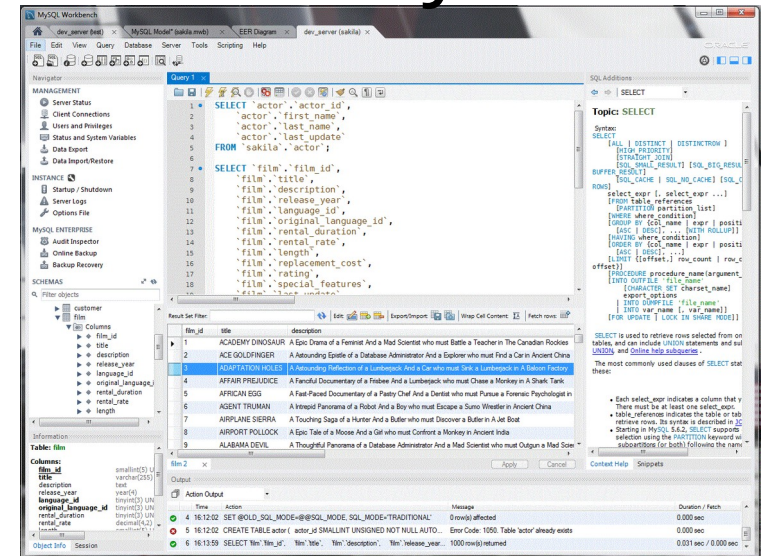
Databases



A structured collection of data stored in a computer system, managed by specialized software (DBMS) for efficient, reliable, and multi-user access

DBMS: Database Management System

How do they look like



Managing large and complex data is difficult

- In the early days, applications to store data were built directly on top of file systems, leading to
 - **Data redundancy and inconsistency**
 - Data is stored in multiple file formats resulting in duplication of information in different files
 - **Difficulty in accessing data**
 - Need to write a new program to carry out each new task
 - **Data isolation**
 - Multiple files and formats
 - **Integrity problems**
 - Integrity constraints (e.g., $\text{account balance} > 0$) become “buried” in program code rather than being stated explicitly
 - Hard to add new constraints or change existing ones

Managing large and complex data is difficult

- **Atomicity of updates**

- Failures may leave database in an inconsistent state with partial updates carried out
- Example: transfer of funds from one account to another should either complete or not happen at all

- **Concurrent access by multiple users**

- Concurrent access needed for performance
- Uncontrolled concurrent accesses can lead to inconsistencies
 - Ex: Two people reading a balance (say 100) and updating it by withdrawing money (say 50 each) at the same time

- **Security problems**

- Hard to provide user access to some, but not all, data

How big? How Complex?

- Walmart's Retail Link database manages its supply chain and inventory management
 - Inventory levels
 - Sales data
 - Supply chain logistics
- Used to
 - Assess product performance
 - Inventory replenishment
 - Pricing strategies
 - Demand forecasting
- Scale:
 - Billions of transactions per day
 - Integrating data from thousands of suppliers and millions of products
- Fun story about Walmart
 - *Strawberry Pop-Tarts increase in sales, like seven times their normal sales rate, ahead of a hurricane. And the pre-hurricane top-selling item was beer*



Is this someway useful for me?

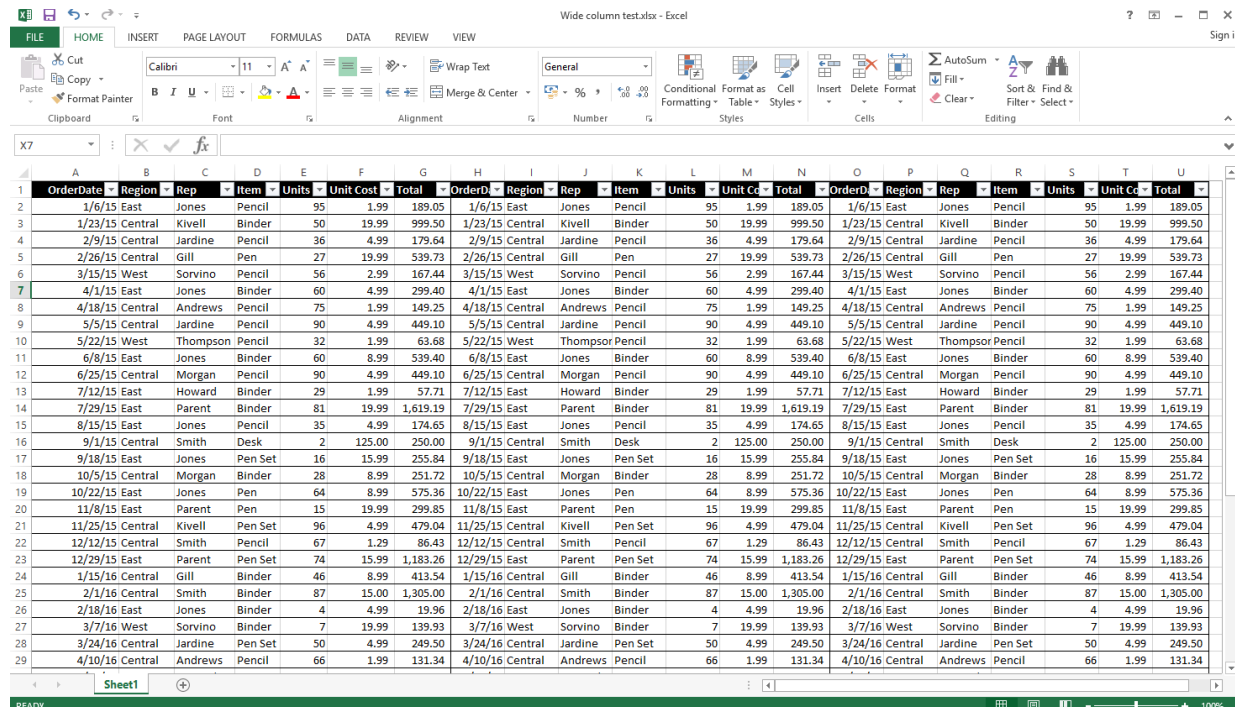
- <https://www.bls.gov/ooh/computer-and-information-technology/data-base-administrators.html>



<https://www.university2business.it/ricerca-del-lavoro/le-7-professioni-sempre-piu-richieste-in-ambito-big-data/>

Quick poll

- How many of you
 - Heard of SQL before
 - Used SQL before
 - Called this “my database”



The screenshot shows an Excel spreadsheet titled "Wide column test.xlsx". The spreadsheet contains multiple tables of data, each with columns for OrderDate, Region, Rep, Item, Units, Unit Cost, and Total. The data is organized into repeating blocks of columns, with some rows highlighted in green. The interface includes the standard Excel ribbon with tabs for FILE, HOME, INSERT, PAGE LAYOUT, FORMULAS, DATA, REVIEW, and VIEW. The status bar at the bottom indicates "READY" and "Sheet1".

OrderDate	Region	Rep	Item	Units	Unit Cost	Total	OrderDate	Region	Rep	Item	Units	Unit Cost	Total	OrderDate	Region	Rep	Item	Units	Unit Cost	Total
1/6/15	East	Jones	Pencil	95	1.99	189.05	1/6/15	East	Jones	Pencil	95	1.99	189.05	1/6/15	East	Jones	Pencil	95	1.99	189.05
1/23/15	Central	Kivell	Binder	50	19.99	999.50	1/23/15	Central	Kivell	Binder	50	19.99	999.50	1/23/15	Central	Kivell	Binder	50	19.99	999.50
2/9/15	Central	Jardine	Pencil	36	4.99	179.64	2/9/15	Central	Jardine	Pencil	36	4.99	179.64	2/9/15	Central	Jardine	Pencil	36	4.99	179.64
2/26/15	Central	Gill	Pen	27	19.99	539.73	2/26/15	Central	Gill	Pen	27	19.99	539.73	2/26/15	Central	Gill	Pen	27	19.99	539.73
3/15/15	West	Sorvino	Pencil	56	2.99	167.44	3/15/15	West	Sorvino	Pencil	56	2.99	167.44	3/15/15	West	Sorvino	Pencil	56	2.99	167.44
4/1/15	East	Jones	Binder	60	4.99	299.40	4/1/15	East	Jones	Binder	60	4.99	299.40	4/1/15	East	Jones	Binder	60	4.99	299.40
4/18/15	Central	Andrews	Pencil	75	1.99	149.25	4/18/15	Central	Andrews	Pencil	75	1.99	149.25	4/18/15	Central	Andrews	Pencil	75	1.99	149.25
5/5/15	Central	Jardine	Pencil	90	4.99	449.10	5/5/15	Central	Jardine	Pencil	90	4.99	449.10	5/5/15	Central	Jardine	Pencil	90	4.99	449.10
5/22/15	West	Thompson	Pencil	32	1.99	63.68	5/22/15	West	Thompson	Pencil	32	1.99	63.68	5/22/15	West	Thompson	Pencil	32	1.99	63.68
6/8/15	East	Jones	Binder	60	8.99	539.40	6/8/15	East	Jones	Binder	60	8.99	539.40	6/8/15	East	Jones	Binder	60	8.99	539.40
6/25/15	Central	Morgan	Pencil	90	4.99	449.10	6/25/15	Central	Morgan	Pencil	90	4.99	449.10	6/25/15	Central	Morgan	Pencil	90	4.99	449.10
7/12/15	East	Howard	Binder	29	1.99	57.71	7/12/15	East	Howard	Binder	29	1.99	57.71	7/12/15	East	Howard	Binder	29	1.99	57.71
7/29/15	East	Parent	Binder	81	19.99	1,619.19	7/29/15	East	Parent	Binder	81	19.99	1,619.19	7/29/15	East	Parent	Binder	81	19.99	1,619.19
8/15/15	East	Jones	Pencil	35	4.99	174.65	8/15/15	East	Jones	Pencil	35	4.99	174.65	8/15/15	East	Jones	Pencil	35	4.99	174.65
9/1/15	Central	Smith	Desk	2	125.00	250.00	9/1/15	Central	Smith	Desk	2	125.00	250.00	9/1/15	Central	Smith	Desk	2	125.00	250.00
9/18/15	East	Jones	Pen Set	16	15.99	255.84	9/18/15	East	Jones	Pen Set	16	15.99	255.84	9/18/15	East	Jones	Pen Set	16	15.99	255.84
10/5/15	Central	Morgan	Binder	28	8.99	251.72	10/5/15	Central	Morgan	Binder	28	8.99	251.72	10/5/15	Central	Morgan	Binder	28	8.99	251.72
10/22/15	East	Jones	Pen	64	8.99	575.36	10/22/15	East	Jones	Pen	64	8.99	575.36	10/22/15	East	Jones	Pen	64	8.99	575.36
11/8/15	East	Parent	Pen	15	19.99	299.85	11/8/15	East	Parent	Pen	15	19.99	299.85	11/8/15	East	Parent	Pen	15	19.99	299.85
11/25/15	Central	Kivell	Pen Set	96	4.99	479.04	11/25/15	Central	Kivell	Pen Set	96	4.99	479.04	11/25/15	Central	Kivell	Pen Set	96	4.99	479.04
12/12/15	Central	Smith	Pencil	67	1.29	86.43	12/12/15	Central	Smith	Pencil	67	1.29	86.43	12/12/15	Central	Smith	Pencil	67	1.29	86.43
12/29/15	East	Parent	Pen Set	74	15.99	1,183.26	12/29/15	East	Parent	Pen Set	74	15.99	1,183.26	12/29/15	East	Parent	Pen Set	74	15.99	1,183.26
1/15/16	Central	Gill	Binder	46	8.99	413.54	1/15/16	Central	Gill	Binder	46	8.99	413.54	1/15/16	Central	Gill	Binder	46	8.99	413.54
2/1/16	Central	Smith	Binder	87	15.00	1,305.00	2/1/16	Central	Smith	Binder	87	15.00	1,305.00	2/1/16	Central	Smith	Binder	87	15.00	1,305.00
2/18/16	East	Jones	Binder	4	4.99	19.96	2/18/16	East	Jones	Binder	4	4.99	19.96	2/18/16	East	Jones	Binder	4	4.99	19.96
3/7/16	West	Sorvino	Binder	7	19.99	139.93	3/7/16	West	Sorvino	Binder	7	19.99	139.93	3/7/16	West	Sorvino	Binder	7	19.99	139.93
3/24/16	Central	Jardine	Pen Set	50	4.99	249.50	3/24/16	Central	Jardine	Pen Set	50	4.99	249.50	3/24/16	Central	Jardine	Pen Set	50	4.99	249.50
4/10/16	Central	Andrews	Pencil	66	1.99	131.34	4/10/16	Central	Andrews	Pencil	66	1.99	131.34	4/10/16	Central	Andrews	Pencil	66	1.99	131.34

How we use databases at Shair.Tech

- As the basic infrastructure of our products (BringTheFood)
- As the data storage for monitoring applications (Analytics and Statistics)
- As the data management systems for business information (Redmine, CRM)
- As an efficient system to store and extract data for reporting purposes

Databases at Shair.Tech

